Project Title : Declining Sales in a Retail Business

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Business Problem Framing Report

Topic : Analysis of Declining Sales in a Retail Business

1. Introduction

A retail business sells products to customers through physical stores or online channels. Sales performance is important for business growth and profitability. Sometimes, a business experiences a decline in sales due to changing customer preferences, product availability, pricing, competition, or seasonal demand. Data analytics can help identify the possible reasons for this decline and support better business decisions.

2. Problem Statement

The retail business has observed a decline in sales revenue over recent months. However, the exact reasons for this decline are not known. The business needs to analyze historical sales data to identify the products, regions, customer segments, or sales channels contributing to the decline.

3. Business Objective

The main objective is to analyze sales data, identify the major factors associated with declining sales, and provide data-driven recommendations to improve sales performance.

4. Key Business Questions

- Which products generate the highest and lowest sales?
- Which regions have experienced a sales decline?
- How have monthly sales changed over time?
- Which sales channels perform best?
- Are discounts associated with changes in sales revenue or profit?
- What actions could help improve sales?

5. Required Data Fields

Data field	Purpose
Order ID	Identify individual orders
Order Date	Analyze monthly and yearly trends
Product Name	Compare product performance
Quantity Sold	Measure sales volume
Unit Price	Understand product pricing
Sales Revenue	Measure total sales
Cost and Profit	Evaluate profitability
Region or City	Compare geographical performance
Sales Channel	Compare online and offline sales
Discount	Analyze the relationship between discounts and sales

6. Possible Data Sources

- Company sales records or billing systems.
- Excel or CSV sales files.
- Point-of-sale (POS) systems.
- E-commerce order records.
- Publicly available retail datasets, such as those on Kaggle.

7. Assumptions

- The dataset contains sufficiently accurate sales records.
- Order dates and sales values are recorded consistently.
- The dataset covers a suitable period for comparison, such as 6–12 months.
- Revenue and profit are calculated consistently.
- Missing or incorrect records will be checked before analysis.

8. Analytics Plan

The analysis will be carried out in the following steps.

1. Data collection : Obtain sales data from company records, Excel/CSV files, or a public dataset.
2. Data cleaning : Check missing values, duplicate orders, incorrect dates, and invalid sales figures.
3. Exploratory data analysis : Calculate total revenue, total profit, quantity sold, and monthly sales trends.
4. Comparative analysis : Compare sales by product, region, customer segment, and sales channel.
5. Visualization : Create charts in Excel or Power BI to identify patterns and areas of concern.
6. Recommendations : Suggest actions based on the results, such as reviewing underperforming products, improving stock availability, or adjusting promotional strategies where supported by evidence.

9. Expected Outcomes

- Identify products and regions with declining sales.
- Understand monthly sales patterns.
- Discover areas requiring further investigation.
- Support decisions using measurable business indicators.
- Provide practical recommendations for improving sales and profitability.

10. Conclusion

Business problem framing provides a clear direction for a data analytics project. By defining the problem, establishing objectives, identifying the required data, and preparing a structured analysis plan, the retail business can investigate declining sales systematically. The actual causes and recommendations will be finalized after analyzing the collected data.